Building identifiable kinetic structures from steady-state data: A case study from *Escherichia coli* glycolysis

1st Gabriel I. Miranda-Contreras
Department of Chemical and Bioprocess Engineering
Pontifical Catholic University of Chile
Santiago, Chile
ggmiranda@uc.cl

2nd María Trinidad Smith Muñoz
Department of Chemical and Bioprocess Engineering
Pontifical Catholic University of Chile
Santiago, Chile
mt.smith@uc.cl

3rd Eloy García
Department of Chemical and Bioprocess Engineering
Pontifical Catholic University of Chile
Santiago, Chile
eloy.garcia@uc.cl

4th José R. Pérez-Correa
Department of Chemical and Bioprocess Engineering
Pontifical Catholic University of Chile
Santiago, Chile
jperezc@uc.cl

5th Pedro Saa
Department of Chemical and Bioprocess Engineering
Institute for Mathematical and Computational Engineering
Pontifical Catholic University of Chile
Santiago, Chile
pnsaa@uc.cl

***Abstract*—Kinetic models provide a mechanistic framework for studying cellular metabolism, capturing enzyme saturation, regulation consistent with thermodynamic constraints that stoichiometric approaches cannot describe. However, their large number of parameters, combined with scarce and noisy experimental data, often renders them practically non-identifiable. The challenge is even greater using steady-state date, since many different parameters can produce similar results. In this work, a mechanistic kinetic model of the central glycolytic pathway of *Escherichia coli* was constructed, comprising 9 enzymatic reactions and 19 metabolites, using the separable thermodynamic rate law of Noor et al. [1] and experimental dataset of Ishii et al. [2]. A systematic parameter-sensitivity-based methodology is proposed to reduce the number of free parameter while preserving predictive performance. Starting from 37 kinetic parameters, by combining FIM-based correlations, normalized sensitivities coefficients, and singular value analysis of the relative sensitivity matrix, a final model with 16 free parameters was obtained. The reduced model improves both metabolite-concentration and flux predictions relative to a purely literature-parametrized model while leave-one-out cross validation confirmed comparable predictive capacity on held-out conditions, particularly when predicting fluxes. Parameter bootstraping and profile-likelihood analyses enabled characterization of the reduced model, revealing multimodal parameter distributions and multiple local minima, conditions that limit gradient-based optimizers such as IPOPT. These results show that reducing the number of parameters from 37 to 16 improves identifiability, but that data scarcity, high experimental variability, and optimizer limitations remain critical for building predictive kinetic models. As more data become available, we foresee kinetic parameter reduction as a necessary step for constructing robust and accurate models.**

***Index Terms*—Metabolic modeling, kinetic models, sensitivity analysis, model selection**

I. Introduction

Living cells continuously transform nutrients into energy and biomass through coordinated biochemical reactions organized into metabolic networks, where a multitude of enzymatic reactions operate simultaneously and share metabolites and cellular resources [3]. Since metabolic function shapes an organism's phenotype and response to genetic or environmental perturbations [4], understanding these networks is central to biotechnology.

Mathematical models of these networks enable to identify rate-limiting steps and anticipate the system effects of genetic or environmental perturbations without exhaustive experiments [5], although the strong nonlinearities, tight regulation and multi-scale interactions of metabolism make construction of such representations challenging [6].

Stoichiometric models represent cellular metabolism through the stoichiometric matrix to identify feasible steady state flux distributions throughout the network [7]. Methods such as Flux Balances Analysis (FBA) enable computation of flux distributions without requiring detailed kinetic information [8], making them suitable for genome-scale analysis [8]. However, because reaction rates are not explicitly linked to metabolite concentrations, enzyme abundance or regulatory mechanisms, these models cannot

capture intracellular transient dynamics or the mechanisms underlying metabolic fluxes [7].

Kinetic models address this gap by incorporating rate laws, describing the dynamic evolution of intracellular metabolites via systems of ordinary differential equations capturing transient responses, enzyme saturation, reversibility, regulation and thermodynamic constraints [1]. This mechanistic detail allows to predict how metabolic states are achieved, rather than only which flux distributions are feasible [4]. However, their construction remains challenging: rate laws are highly nonlinear, the parameter space is large, and data are scarce. Since different parameter combinations can reproduce similar behaviour, parameter identifiability remains one of the central open problems in the field [5].

This work proposes a novel methodology for building kinetic models using parameter sensitivity analyses. A a case study, the central glycolytic pathway of *Escherichia coli*, the fundamental route converting glucose into pyruvate while generating ATP was analyzed. This is among the most extensively studied metabolic pathways, owing to the availability of metabolomic, fluxomic, and enzymatic data that make it an attractive benchmark for developing and validating kinetic models [2]. This work constructed a kinetic model of this pathway using the thermodynamic rate law of Noor et al. [1]. Through systematic model reduction based on sensitivity analyses, a reduced model structure that retains flux prediction capabilities was generated and then evaluated by characterizing parameter uncertainty using profile likelihood, and assessing predictive capacity via cross-validation. Together, these analyses yielded a reduced model structure and identified future work to strengthen the kinetic modelling of metabolic pathways.

II. Methodology

A. Mechanistic kinetic models

Kinetic models describe the metabolism of a species through a system of ordinary differential equations (ODEs). This system describes the evolution of intracellular metabolites with the following general expression:

$$\frac{d\mathbf{x}}{dt} = \mathbf{S} \cdot \mathbf{v}(\mathbf{e}, \mathbf{x}, \mathbf{k}), \quad \mathbf{x}(0) = \mathbf{x}_0 \tag{1}$$

where $\mathbf{S}$ is the stoichiometric matrix, $\mathbf{v}$ is the vector of reaction rates, $\mathbf{e}$ is the enzyme concentration vector, $\mathbf{x}$ is the vector of metabolite concentrations, and $\mathbf{k}$ collects all kinetic parameters [4]. Under the assumption of steady state, i.e., $\mathbf{S} \cdot \mathbf{v} = \mathbf{0}$, the system yields a stationary flux distribution that characterizes the metabolic phenotype of the cell [8].

Different mathematical formulations can be defined to model the dynamics of metabolites in the cell defined by Eqn. (1). Particularly, the thermodynamic rate law presented by Noor et al. [1] is adopted due to its simplicity and interpretability. This rate law takes the form:

$$v = V^+ \cdot \kappa \cdot \gamma \tag{2}$$

where V^+ denotes an enzyme capacity term, κ is the fractional saturation term, and γ describes a thermodynamic driving force term. These are defined as:

$$V^+ = e \cdot k^+_{cat} \tag{3}$$

$$\kappa = \frac{\prod_{i \in \mathcal{I}} \left(\frac{[s_i]}{K_{s,i}} \right)^{m_i}}{1 + \prod_{i \in \mathcal{I}} \left(\frac{[s_i]}{K_{s,i}} \right)^{m_i} + \prod_{j \in \mathcal{J}} \left(\frac{[p_j]}{K_{p,j}} \right)^{m_j}} \tag{4}$$

$$\gamma = 1 - \frac{1}{K'_{eq}} \frac{\prod_{j \in \mathcal{J}} [p_j]^{m_j}}{\prod_{i \in \mathcal{I}} [s_i]^{m_i}} \tag{5}$$

where e is the enzyme concentration, k^+_{cat} is the forward catalytic constant, $[s_i]$ and $[p_j]$ are, respectively, the concentrations of substrate i and product j, $K_{s,i}$ and $K_{p,j}$ are their respective Michaelis-Menten constants, m_i and m_j are the stoichiometric coefficients of substrate i and product j in the reaction, $\mathcal{I}$ and $\mathcal{J}$ denote the index sets of substrates and products and K'_{eq} is the equilibrium constant [1].

B. Kinetic model of E. coli's glycolisis

The previous mathematical formulation was applied to a kinetic model of the core glycolytic pathway of *E. coli* S1, comprising 9 enzymatic reactions and 19 metabolites (9 balanced) (38)[9].

The model spans glucose uptake through phosphoenolpyruvate formation and was parameterized using data from EcoCyc [10] and eQuilibrator [11] databases, as well as the experimental data set of Ishii et al. [2].

The data utilized to fit the model consists of a steady-state omic data from *E. coli K-12*. The conditions in the dataset comprise 24 single-gene deletions and four growth conditions under different dilution rates in chemostats. Internal metabolite and enzyme concentration, together with metabolic flux measurements, are reported in each condition. After a data preprocessing step, the dataset comprised 22 experimental conditions composed of 21 mutant strains and 1 wild type cultivation condition.

Reported parameters are summarized in Table-III. The K'_{eq} were computed using eQuilibrator's API imitating cellular conditions (pH = 7, pMg = 3, ionic strength = 0.1). All reactions are formulated using equation (2), with the exception of the phosphotransferase system (PTS), which follows Kadir et al. [12] formulation 20. The resulting rate equations, along with the notation used throughout, can be found in the supplementary section (S-I-A).

C. Steady-state simulation

Computation of the steady-state condition Eqn. (2) was achieved by solving the following optimization problem:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{u}} \quad & 0 \\ \text{s.t.} \quad & \mathbf{Sv}(\mathbf{x},\mathbf{u},\mathbf{e};\mathbf{k}) - \mathbf{b} = \mathbf{0} \\ & \mathbf{x_{lb}} \leq \mathbf{x} \leq \mathbf{x_{ub}} \\ & \mathbf{u_{lb}} \leq \mathbf{u} \leq \mathbf{u_{ub}} \\ & \mathbf{x} \in \mathbb{R}^{n_{eq}}, \mathbf{u} \in \mathbb{R}^{n_{neq}} \end{aligned} \tag{6}$$

where $\mathbf{e}$, $\mathbf{x}$, $\mathbf{u}$ and $\mathbf{k}$ are the vectors of enzyme concentration, balanced metabolite concentration, imbalanced metabolite concentration and kinetic parameters, respectively. This problem is a constraint satisfaction problem, where the objective is to determine the set of metabolites that satisfy the steady-state condition within the feasible region [13]. The distinction between balanced and imbalanced metabolites is that the first are encoded into the stoichiometric matrix that represents the system (see Eqn. (38)), meanwhile the latter represent metabolites that are in abundance in the cell. The $\mathbf{b}$ vector represents metabolic drains associated to biomass formation not considered in the pathway.

D. Model reduction

1) Parameter estimation: The reduction of a model consists on reducing the amount of parameters to estimate. For this task, the following parameter estimation problem is described. The vector of original parameters $\mathbf{k}$ can be re-defined as $\mathbf{k} = [\boldsymbol{\theta}; \mathbf{k_f}]$, where $\mathbf{k_f}$ are the fixed parameters and $\boldsymbol{\theta}$ are the free parameters. To estimate $\boldsymbol{\theta}$, the following non-linear problem is formulated:

$$\min_{\{\mathbf{x}_e,\mathbf{u}_e\}_{e\in\mathcal{E}},\,\boldsymbol{\theta}} \quad g(\mathbf{v},\hat{\mathbf{v}},\mathbf{x},\hat{\mathbf{x}};\theta) \tag{7}$$

$$\text{s.t.} \quad \left.\begin{aligned} & \mathbf{S}\,\mathbf{v}(\mathbf{x}_e,\mathbf{u}_e,\mathbf{e}_e;\mathbf{k_f},\boldsymbol{\theta}) - \mathbf{b}_e = \mathbf{0} \\ & \mathbf{x_{lb}} \leq \mathbf{x}_e \leq \mathbf{x_{ub}} \\ & \mathbf{u_{lb}} \leq \mathbf{u}_e \leq \mathbf{u_{ub}} \\ & \mathbf{x}_e \in \mathbb{R}^{n_{eq}},\ \mathbf{u}_e \in \mathbb{R}^{n_{neq}} \end{aligned}\right\} \forall e \in \mathcal{E}$$

$$\boldsymbol{\theta_{lb}} \leq \boldsymbol{\theta} \leq \boldsymbol{\theta_{ub}}, \quad \boldsymbol{\theta} \in \mathbb{R}^{n_\theta}$$

with

$$g(\mathbf{v},\hat{\mathbf{v}},\mathbf{x},\hat{\mathbf{x}};\theta) = \sum_{e\in\mathcal{E}} (WSSE_{v,e} + WSSE_{v,e}) \tag{8}$$

$$WSSE_{v,e} = (\mathbf{v}_e - \hat{\mathbf{v}}_e)^T \mathbf{Q_v} (\mathbf{v}_e - \hat{\mathbf{v}}_e) \tag{9}$$

$$WSSE_{x,e} = (\mathbf{x}_e - \hat{\mathbf{x}}_e)^T \mathbf{Q_x} (\mathbf{x}_e - \hat{\mathbf{x}}_e) \tag{10}$$

where $\mathbf{v}_{real,e}$ and $\mathbf{x}_{real,e}$ denote the measured fluxes and metabolite concentrations under experimental condition e. The $\mathbf{Q_v}$ and $\mathbf{Q_x}$ matrices contain the measurement error which is determined by the experimental replicate standard deviation.

2) Parameter identifiability analysis: A critical concern in kinetic modelling is the reliability of the estimated parameters. The goal is to find a set of parameters that can be uniquely estimated from experimental data, i.e., identifiable. This notion splits into structural and practical identifiability, where the former is a property of the model itself,assuming noise-free data, while the latter depends on the size and quality of the data [14]. The methods described in this section were used to assess the reliability of the model parameters. Based on this analysis, highly correlated parameters with low influence on the model states were fixed, thereby reducing the number of free parameters and improving their identifiability.

a) Correlation and sensitivity analysis: The parameter sensitivity coefficients are defined as:

$$G_{ij} = \frac{\partial x_i}{\partial \theta_j} \tag{11}$$

Considering the steady-state condition:

$$f(\mathbf{x},\mathbf{u},\mathbf{e},\mathbf{k}) = \mathbf{Sv}(\mathbf{x},\mathbf{u},\mathbf{e};\mathbf{k}) - \mathbf{b} = \mathbf{0} \tag{12}$$

the relative sensitivities can be computed as:

$$\frac{\partial \mathbf{x}}{\partial \theta_j} = -\left(\mathbf{S}\frac{\partial \mathbf{v}}{\partial \mathbf{x}}\right)^{-1} \mathbf{S}\frac{\partial \mathbf{v}}{\partial \theta_j}\bigg|_{\mathbf{x}} \tag{13}$$

$$\frac{\partial \mathbf{v}}{\partial \theta_j} = \frac{\partial \mathbf{v}}{\partial \mathbf{x}}\frac{\partial \mathbf{x}}{\partial \theta_j} + \frac{\partial \mathbf{v}}{\partial \theta_j}\bigg|_{\mathbf{x}} \tag{14}$$

With this formulation, the Fisher Information Matrix (FIM) can be constructed as:

$$\mathbf{F} = \sum_{i=1}^{m} \mathbf{G_i^T} \mathbf{Q}^{-1} \mathbf{G}_i \tag{15}$$

where m are the different experimental conditions and $\mathbf{Q}$ corresponds to the measurement error matrix. The FIM can then be used to construct the parameter covariance matrix (Σ)[15]:

$$\mathbf{F^{-1}} = \boldsymbol{\Sigma} \tag{16}$$

Furthermore, the correlation coefficient between parameters may be estimated from the covariance matrix through the following expression:

$$r_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j} \tag{17}$$

Where $\sigma_{ij} = \Sigma_{ij}$ is the covariance between the i-th parameter and the j-th parameter, and $\sigma_i = \sqrt{\Sigma_{ii}}$ is the standard deviation of the i-th parameter. This metric provides insights into which parameter should be fixed by identifying highly correlated parameters, e.g., $|r_{ij}| > 0.9$.

Besides the correlation analysis, the singular value decomposition of the relative sensitivity coefficient ($\overline{G}_{ij} = \theta_j / x_i \cdot G_{ij}$) provides information about parameter correlations with nonsignificant information [16]. In particular, the $\overline{G}_{ij}$ matrix can be decomposed as:

$$\overline{G}_{ij} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + ... + \sigma_p \mathbf{u}_p \mathbf{v}_p^T \tag{18}$$

following its Singular Value Decomposition (SVD). Here the eigenvectors v_i contain information about contribution of the parameters to that sensitivity direction. This contribution is proportional to the related eigenvalues σ_i. Small values σ_i imply that the parameters with non-zero components in v_i are correlated and consequently are candidates to be fixed.

E. Model Characterization

After the model reduction, a characterization of the parameters and model performance can be done. Different methodologies can be applied for this task, in particular the profile likelihood, boostrapping and Leave-One-Out cross validation (LOOCV) methods were used.

1) Profile likelihood: The impact of a parameter θ_i on the model fit can be quantified through its profile likelihood, defined as:

$$g_{profile}(\theta_i) = \min_{\boldsymbol{\theta}_{-i}} g(\mathbf{v}, \hat{\mathbf{v}}, \mathbf{x}, \hat{\mathbf{x}}; \theta_i, \boldsymbol{\theta}_{-i}) \tag{19}$$

where $\boldsymbol{\theta}_{-i}$ is the set of parameters excluding θ_i. Evaluating this profile over a range of values of θ_i reveals the sensitivity of the objective function to that parameter, and thus its identifiability [14]. A flat profile indicates structural non-identifiability, while a plateau below the threshold: $\Delta(\alpha) = icdf(\chi^2_1, \alpha)$, indicates practical non-identifiability, as the data provide little information to constrain the parameter; otherwise, the parameter is identifiable [14].

2) Empirical parameter uncertainty quantification (bootstrapping): The covariance matrix presented at Eqn. 16 is an approximation to the parameter uncertainty and depends on the singularity of the $\mathbf{F}$ matrix. The empirical parameter uncertainty can instead be constructed through the bootstrapping procedure described in Justus et al. [17], which consists in solving the problem described at Eqn. 7 with different sets of experiments resampled from the true data set. This generates the empirical distribution of adjusted parameters ($N = 5000$), allowing the uncertainty of the parameters to be quantified without the stringent assumptions of the FIM.

3) LOOCV: The LOOCV procedure allows to estimate the predictive capacity a the model. This analysis consists of estimating the parameters under all experimental conditions except one and using the estimated parameters to predict the held-out conditions. The defines whether the model is able to estimate conditions outside its training dataset.

F. Computational implementation

Kinetic models were implemented in `Python`, version 3.12.9. The code base and supplementary information associated with the project can be found in the following GitHub repository: https://github.com/ggmirandac/HybridKinetics-IIQ3733.

To solve Eqn. 6 , the Interior Point OPTimizer (IPOPT)[18] was used within the Pyomo package [19]. More specifically, the `parmest` method was used to estimate the parameters and multi-start initialization was used to avoid the dependency on the initial guess of the optimization.

III. Results and Discussion

A. Systematic parameter reduction

Model reduction was carried out as a two-stage, FIM-based procedure applied to two independent starting points, so that the analysis was structural and practical.

a) Starting points: Two parameter sets were used to initialize the reduction pipeline. The first corresponded to the *literature parameter estimation*, in which all 37 kinetic parameters were fixed to the values reported in EcoCyc, eQuilibrator and Bar-Even et al. [20] without any data-driven adjustment (Figure S2). The second corresponds to the *initial parameter estimation*, in which the same 37 parameters were left free and fitted against the Ishii et al. [2] dataset by solving Eqn. 7 (Figure S3). As summarized in Figure 1, the literature-only parametrization performs markedly worse than the data-driven one, particularly on fluxes ($\text{RMSE}_{flux} = 3.905$, $R^2_{flux} = -6.063$), confirming that reported literature values alone are insufficient to reproduce the flux distribution measured under the experimental conditions of Ishii et al. [2].

b) First reduction step: Each of the two starting points was independently analysed using the normalized sensitivity coefficients and the FIM. From the literature-fixed set, only the FIM-derived parameter correlations were computed, since no free parameters were estimated. From the initial free-parameter estimation, normalized sensitivities and FIM-derivated correlations and confidence intervals were jointly used to flag parameters that were both highly correlated ($|r_{ij}| > 0.9$) and weakly influential on the model outputs (Figure S4). The catalytic constants k_{cat} were intentionally left free, as they provide flexibility for model calibration. Combining both analyses, 19 parameters were fixed by trial and error, yielding the first reduced model with 18 free parameters. This model improves the metabolite-concentration fit over both starting points ($\text{RMSE}_{metabolites} = 0.278$) while preserving the flux fit ($\text{RMSE}_{flux} = 1.213$, $R^2_{flux} = 0.318$), suggesting that the flux predictions were not sensitive to the fixed parameters.

c) Second reduction step: The first reduced model was then subjected to a second, stricter identifiability screening, keeping the previous methods while incorporating a SVD of the relative sensitivity matrix. This criterion led to fixing 2 additional parameters, producing the final reduced model with 16 free parameters (Figure S5). This second reduction left the flux metrics unchanged and the metabolite metrics essentially unaffected ($\text{RMSE}_{metabolites} = 0.291$ vs. 0.278), showing that the two additionally fixed parameters were structurally redundant rather than informative. The fixed parameter values are presented in Figure S7.

Across all four models $R^2_{metabolites}$ remained negative (Figure S6), so even after reduction the metabolite-concentration predictions are less accurate than simply predicting the experimental mean; this contrasts with the flux predictions, where $R^2_{flux} = 0.318$ is maintained throughout the reduction pipeline. This asymmetry suggests that flux-carrying capacity is well constrained by the fixed stoichiometry and estimated k^+_{cat} values, while steady-state metabolite concentrations remain more sensitive to the parameters fixed during reduction.

B. Reduced model characterization

After the model reduction procedure, a characterization of the model was performed to understand the different pathologies that arise solving Eqn. 6) which prevented a better model

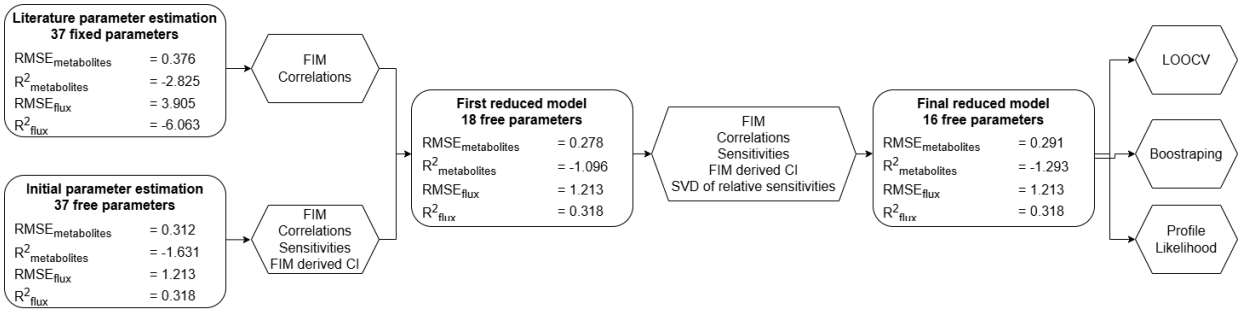


Figure 1: **General workflow for building a reduced kinetic model structure**. The model reduction procedure consisted in the sequential fixation of parameters. The values of the fixed parameters were chosen to the estimated values in the previous step. The model performance metrics, root mean squared error (RMSE) and determination coefficient (R^2), are presented in the rounded squares, meanwhile the methods and metrics computed for each model are mentioned in the hexagonal box.

fit. The model's prediction of metabolites are sensitive to all the free parameters in the model, while having a worse fit compared to the fluxes (Figure 2). This in turn becomes a paradox as the fluxes are not sensitive to the parameters (Figure S5b).

The model parameter values do not change greatly during the estimation procedure, which provides stability in the estimation (Figure 2b). More importantly, the estimated values vary greatly with respect to the literature values which is expected as were measured under different experimental.

When comparing the initial 37 parameter model to the final 16 parameter reduced model, several improvements can be noted. First of all, the RMSE of the metabolites dropped from 0.38 to 0.29 and R^2 improved from -2.82 to -1.29. Secondly, the RMSE of the fluxes dropped from 3.9 to 1.2 and R^2 improved from -6.06 to 0.32 (Figure 2c-d). The LOOCV informs that the reduced model is able to predict outside the training data.

Considering the relatively poor predictive performance, a thorough characterization was carried out. Firstly, an uncertainty quantification was performed, using the the bootstrap distribution of the parameters and objective value. The confidence coefficient coefficient across the different estimations, on average, did not change, remarking that the reduction of the model did not improve the uncertainty over the parameters (Figure 2 e). The bootstrap distributions revealed different issues, chief amongst was the estimated parameter were outside the 95% confidence interval and cases with multiple regions of high probability density (one example can be seen in Figure 2 f). Furthermore, the distribution of the objective value of Eqn. 6 revealed multimodal distributions, together with a wide range of possible objective values. These parameter distribution characteristics hinder the ability of local optimizers, like IPOPT, to find the global minimum across the landscape of local minima.

To asses the latter, the profile likelihood analysis was used to both study the sensitivity of the objective value and optimization to parameter changes, and whether or not are local minima in the vicinity of the estimated parameter. The profile likelihood of the free parameters demonstrated the existence of multiple local minima and artifacts of the optimization method used. For instance, the example shown in Figure 2h shows these two issues. First, the profile showed multiple local minima in the vicinity which are in a basin, hindering gradient-based optimizers performance. Secondly, the optimizer for the same parameter value and initial state found a different objective value, the objective evaluated at the same $\hat{\theta}$ then provides two different values.

IV. Conclusion

Estimating all 37 parameters of the kinetic model proved to be computationally and practically challenging. In order to reduce the model and obtain a better estimation, correlation and sensitivity analyses were performed. With the information provided by these matrices, highly correlated parameters with low influence in the model were fixed, as shown in Figure S7. The model parameters were reduced from 37 to 16. This reduction provided a more identifiable model.

The final reduced model retained predictive capability outside the training data (Figure 2 c-d), but the predictive capabilities over the metabolite set remain poorer in comparison to the fluxes. One reason behind this is the multiplicity of solutions to Eqn. 1, where many concentration sets provide the same steady state solution. Furthermore, the bootstrapping analysis and profile likelihood provided more insights into these pathologies. From the first, the lack of reduction of uncertainty, multimodal parameter and objective-value distributions showed the degeneracy of the problem. The latter finding was further confirmed by the profile likelihood analysis, which clearly showed the presence of multiple local minima.

The issues found may result from omitting regulatory mechanisms in Eqn. 2, which could better capture the complexity of this metabolic pathway; models including these mechanisms are therefore attractive for future implementations. Furthermore, the lack of a dataset large enough to inform the optimization hinders the use of models with many parameters: the original model contained 37 parameters against only 22 experimental conditions. Finally, the use of local optimizers such as IPOPT undermines parameter estimation in a problem with multiple local minima.

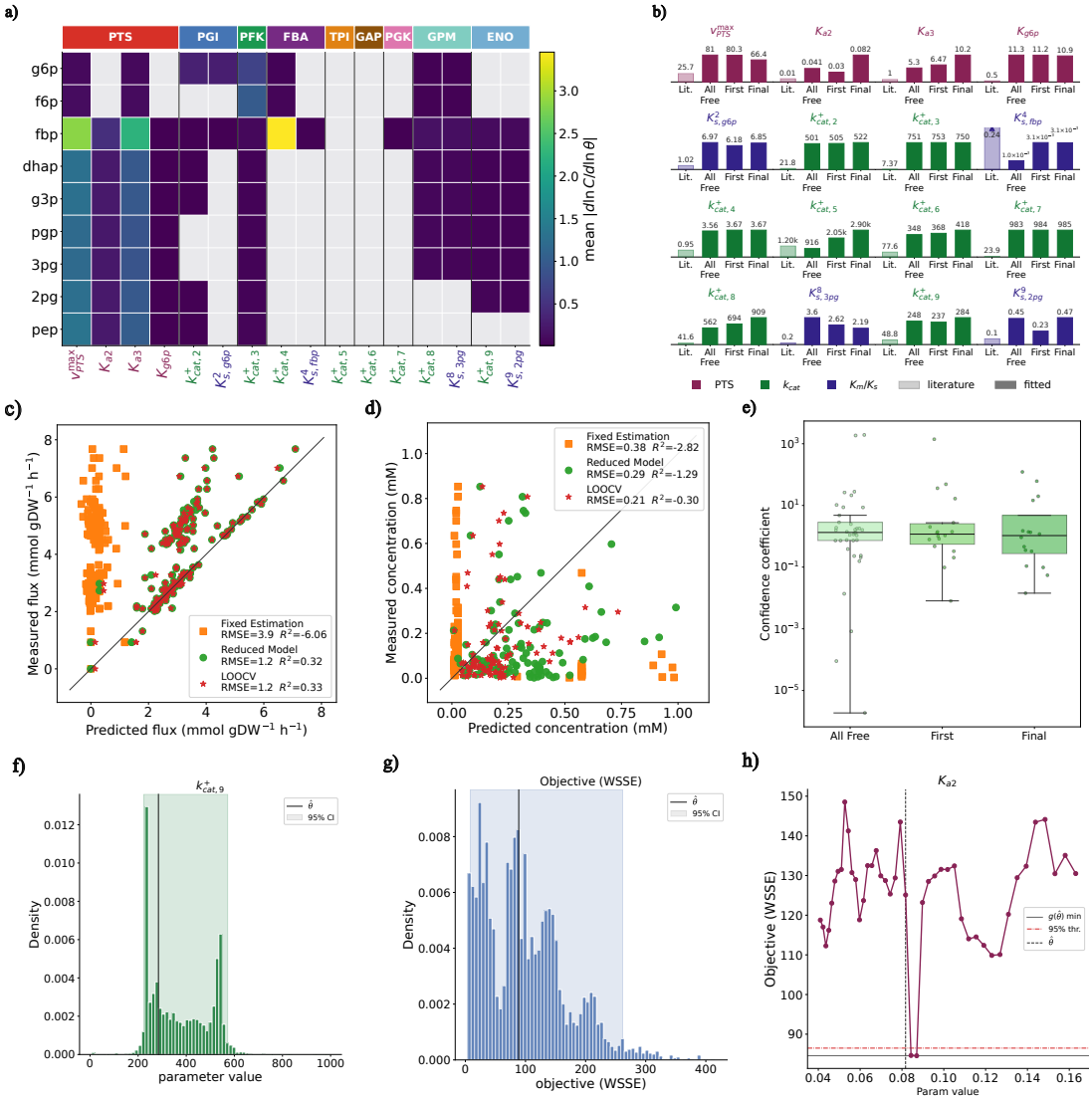


Figure 2: **Reduced model characterization**. Diferent methods are employed to characterize the sensitivity, parameter estimation procedure, performance and parameter uncertainty. **a)** Mean relative sensitivity analysis of the 16 free parameters that span the 9 reactions analysed over the 9 balanced metabolites, the full table can be found in Figure S5b. **b)** Evolution of the 16 free parameters over the course of the optimization and parameter reduction process. Each bar represents the value of the parameter at each step. **c)-d)** Parity plot of model predictions. The model's predictive capability is evaluated with the literature parameters, reduced model fitted parameters and LOOCV analysis. **e)** Objective value distribution obtained through bootstrapping (N=5000). **f)** Examples of problematic parameter distributions, in particular bi-modal distributions, obtained through bootstrapping (N=5000), the complete parameter distributions are shown in Figure S8. **g)** Objective function bootstrap distribution showing multiple zones of high probability density **h)** Example of the profile likelihood of a parameter with the presence of multiple local minima in the Eqn. 6. The complete collection of parameters' profile likelihood can be found in Figure S9.

Acknowledging the limitations of this study, this work lays the groundwork for future efforts in kinetic modelling. Before integration into other pipelines, a comparison with baseline techniques, such as Flux Balance Analysis, must be performed. In addition, the parameter sensitivity problem identified here could be addressed through dynamic data simulations; the results of such would indicate whether dynamic data must be measured to obtain an identifiable model. Finally, global optimization algorithms should be employed in future work to tackle the multiple-local-optima issue, making the optimization more robust. Ultimately, obtaining an identifiable reduced model enables integration with other modelling frameworks, such as hybrid modelling, and with predictive tools like Bayesian parameter estimation over the reduced model.

V. Acknowledgement

GIMC and PS acknowledge the financial support from ANID FONDECYT Regular No. 1262500. GIMC acknowledges the financial support of the National Center for Artificial Inteligence CENIA FB210017, Basal ANID.

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S-I. SUPPLEMENTARY SECTION

A. Kinetic model expressions

Phosphotransferase system (PTS)

$$v_1 = \frac{v_{PTS}^{\max}\,[glc]\,\dfrac{[pep]}{[pyr]}}{K_{a1} + K_{a2}\dfrac{[pep]}{[pyr]} + K_{a3}[glc] + [glc]\,\dfrac{[pep]}{[pyr]}} \cdot \frac{1}{1 + \left(\dfrac{[g6p]^{n_{g6p}}}{K_{g6p}}\right)} \tag{20}$$

Empirical formulation from Kadir et al. [12]; thermodynamic driving force not applicable.

Glucose 6-phosphate isomerase (PGI) EC: 5.3.1.9 $g6p \rightleftharpoons f6p$

$$\kappa_2 = \frac{[g6p]/K_{s,g6p}^2}{1 + [g6p]/K_{s,g6p}^2 + [f6p]/K_{p,f6p}^2} \qquad \gamma_2 = 1 - \frac{[f6p]}{[g6p]\,K'_{eq,2}} \tag{21}$$

$$v_2 = Pgi \cdot k_{cat,2}^{+} \cdot \kappa_2 \cdot \gamma_2 \tag{22}$$

Phosphofructokinase (PFK) EC: 2.7.1.11 $f6p + atp \rightleftharpoons fbp + adp$

$$\kappa_3 = \frac{([f6p]/K_{s,f6p}^3)\,([atp]/K_{s,atp}^3)}{1 + ([f6p]/K_{s,f6p}^3)([atp]/K_{s,atp}^3) + ([fbp]/K_{p,fbp}^3)([adp]/K_{p,adp}^3)} \qquad \gamma_3 = 1 - \frac{[fbp]\,[adp]}{[f6p]\,[atp]\,K'_{eq,3}} \tag{23}$$

$$v_3 = PfkB \cdot k_{cat,3}^{+} \cdot \kappa_3 \cdot \gamma_3 \tag{24}$$

Fructose-bisphosphate aldolase (FBA) EC: 4.1.2.13 $fbp \rightleftharpoons g3p + dhap$

$$\kappa_4 = \frac{[fbp]/K_{s,fbp}^4}{1 + [fbp]/K_{s,fbp}^4 + ([g3p]/K_{p,g3p}^4)([dhap]/K_{p,dhap}^4)} \qquad \gamma_4 = 1 - \frac{[g3p]\,[dhap]}{[fbp]\,K'_{eq,4}} \tag{25}$$

$$v_4 = FbaA \cdot k_{cat,4}^{+} \cdot \kappa_4 \cdot \gamma_4 \tag{26}$$

Triosephosphate isomerase (TPI) EC: 5.3.1.1 $dhap \rightleftharpoons g3p$

$$\kappa_5 = \frac{[dhap]/K_{s,dhap}^5}{1 + [dhap]/K_{s,dhap}^5 + [g3p]/K_{p,g3p}^5} \qquad \gamma_5 = 1 - \frac{[g3p]}{[dhap]\,K'_{eq,5}} \tag{27}$$

$$v_5 = TpiA \cdot k_{cat,5}^{+} \cdot \kappa_5 \cdot \gamma_5 \tag{28}$$

Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) EC: 1.2.1.12 $g3p + pi + nad \rightleftharpoons pgp + nadh$

$$\kappa_6 = \frac{([g3p]/K_{s,g3p}^6)([pi]/K_{s,pi}^6)([nad]/K_{s,nad}^6)}{1 + ([g3p]/K_{s,g3p}^6)([pi]/K_{s,pi}^6)([nad]/K_{s,nad}^6) + ([pgp]/K_{p,pgp}^6)([nadh]/K_{p,nadh}^6)} \tag{29}$$

$$\gamma_6 = 1 - \frac{[pgp]\,[nadh]}{[g3p]\,[pi]\,[nad]\,K'_{eq,6}} \tag{30}$$

$$v_6 = GapA \cdot k_{cat,6}^{+} \cdot \kappa_6 \cdot \gamma_6 \tag{31}$$

Phosphoglycerate kinase (PGK) EC: 2.7.2.3 $pgp + adp \rightleftharpoons 3pg + atp$

$$\kappa_7 = \frac{([pgp]/K_{s,pgp}^7)([adp]/K_{s,adp}^7)}{1 + ([pgp]/K_{s,pgp}^7)([adp]/K_{s,adp}^7) + ([3pg]/K_{p,3pg}^7)([atp]/K_{p,atp}^7)} \qquad \gamma_7 = 1 - \frac{[3pg]\,[atp]}{[pgp]\,[adp]\,K'_{eq,7}} \tag{32}$$

$$v_7 = Pgk \cdot k_{cat,7}^{+} \cdot \kappa_7 \cdot \gamma_7 \tag{33}$$

Phosphoglycerate mutase (GPM) EC: 5.4.2.11 $3pg \rightleftharpoons 2pg$

$$\kappa_8 = \frac{[3pg]/K_{s,3pg}^8}{1 + [3pg]/K_{s,3pg}^8 + [2pg]/K_{p,2pg}^8} \qquad \gamma_8 = 1 - \frac{[2pg]}{[3pg]\,K'_{eq,8}} \tag{34}$$

$$v_8 = GpmA \cdot k_{cat,8}^{+} \cdot \kappa_8 \cdot \gamma_8 \tag{35}$$

Enolase (ENO) EC: 4.2.1.11 $2pg \rightleftharpoons pep$

$$\kappa_9 = \frac{[2pg]/K^9_{s,2pg}}{1 + [2pg]/K^9_{s,2pg} + [pep]/K^9_{p,pep}} \qquad \gamma_9 = 1 - \frac{[pep]}{[2pg]\, K'_{eq,9}} \tag{36}$$

$$v_9 = Eno \cdot k^+_{cat,9} \cdot \kappa_9 \cdot \gamma_9 \tag{37}$$

The stoichiometric matrix of this system is:

$$\mathbf{S} = \begin{array}{c} \\ \text{2pg} \\ \text{3pg} \\ \text{dhap} \\ \text{f6p} \\ \text{fbp} \\ \text{g3p} \\ \text{g6p} \\ \text{pep} \\ \text{pgp} \end{array} \begin{array}{c} \begin{array}{ccccccccc} \text{pts} & \text{pgi} & \text{pfk} & \text{fba} & \text{tpi} & \text{gap} & \text{pgk} & \text{gpm} & \text{eno} \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \end{bmatrix} \end{array} \tag{38}$$

S-II. FIGURES

A. *Escherichia coli* metabolism

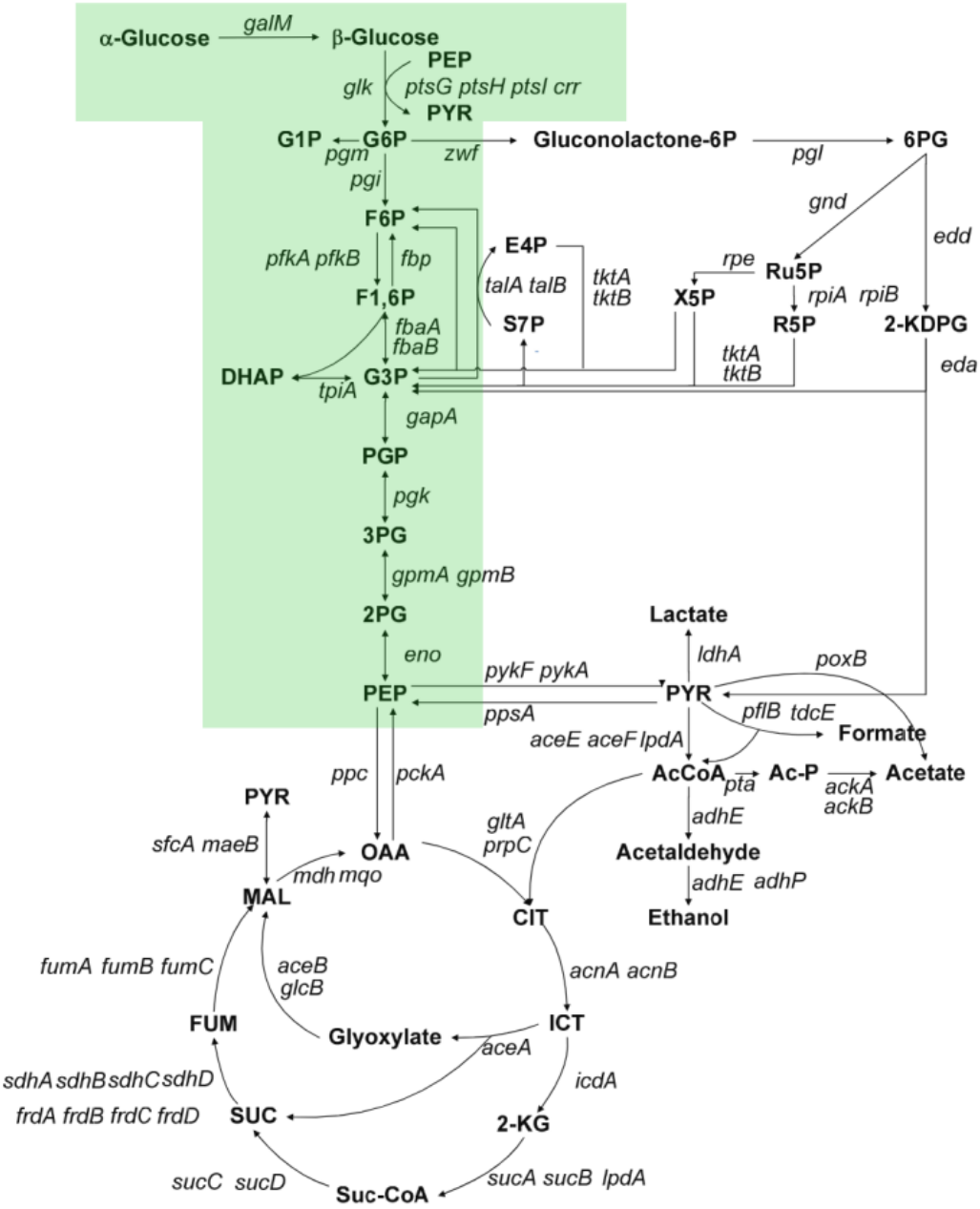


Figure S1: Carbon metabolism pathways of *E.coli* [2]. The reaction highlighted in green encompass the studied reactions.

B. Model descriptions

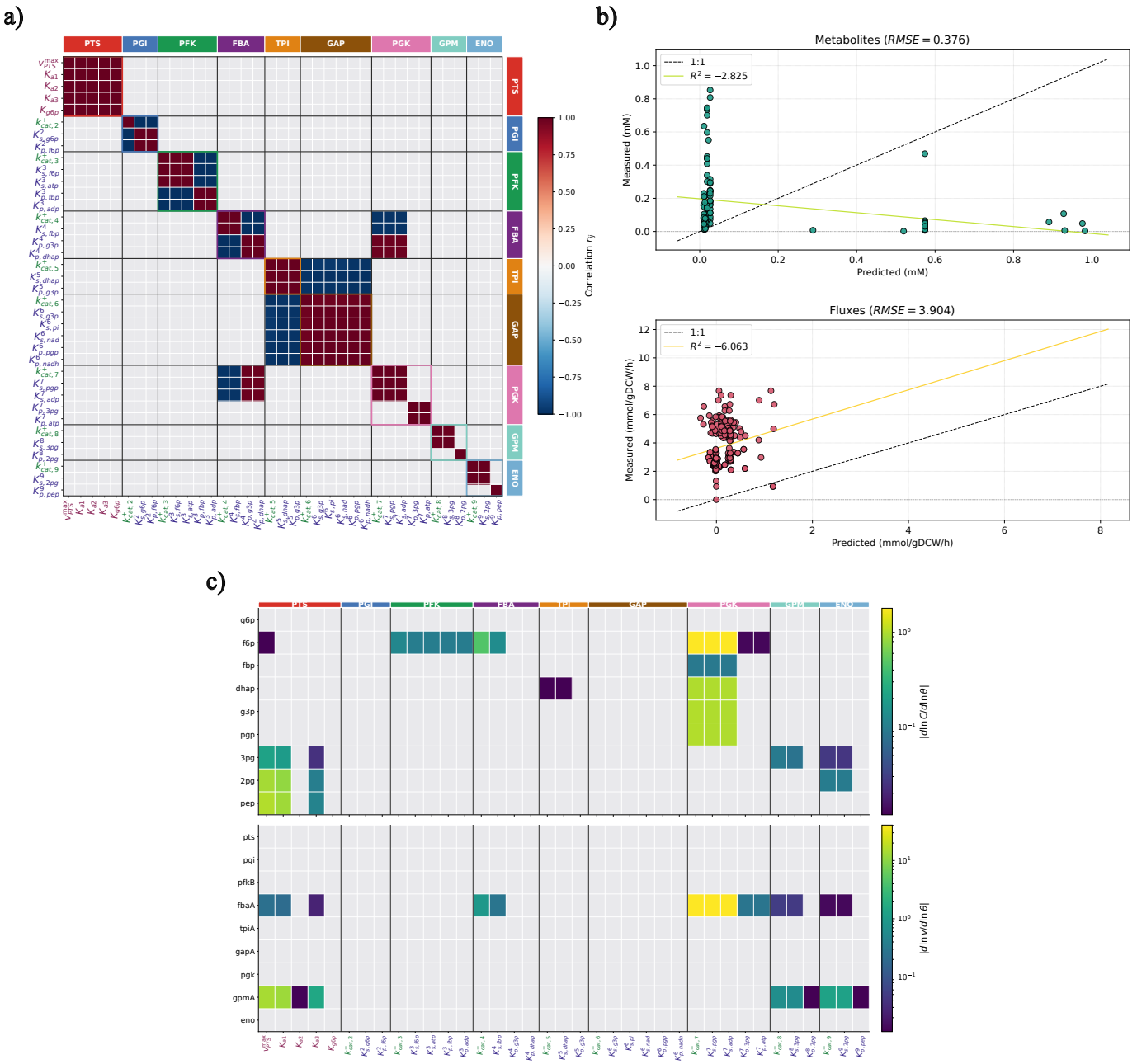


Figure S2: **Fixed-parameters (literature model) analysis**. **a)** FIM-derived parameter correlation matrix computed from the literature-fixed parameter set, grouped by reaction. **b)** Parity plots comparing measured vs. predicted metabolite concentrations (top, RMSE $= 0.376$, $R^2 = -2.825$) and fluxes (bottom, RMSE $= 3.904$, $R^2 = -6.063$) using literature parameter values without any data-driven fitting. **c** Relative sensitivity matrix of the model output to the 37 parameters of the model evaluated at the parameter value and model outputs.

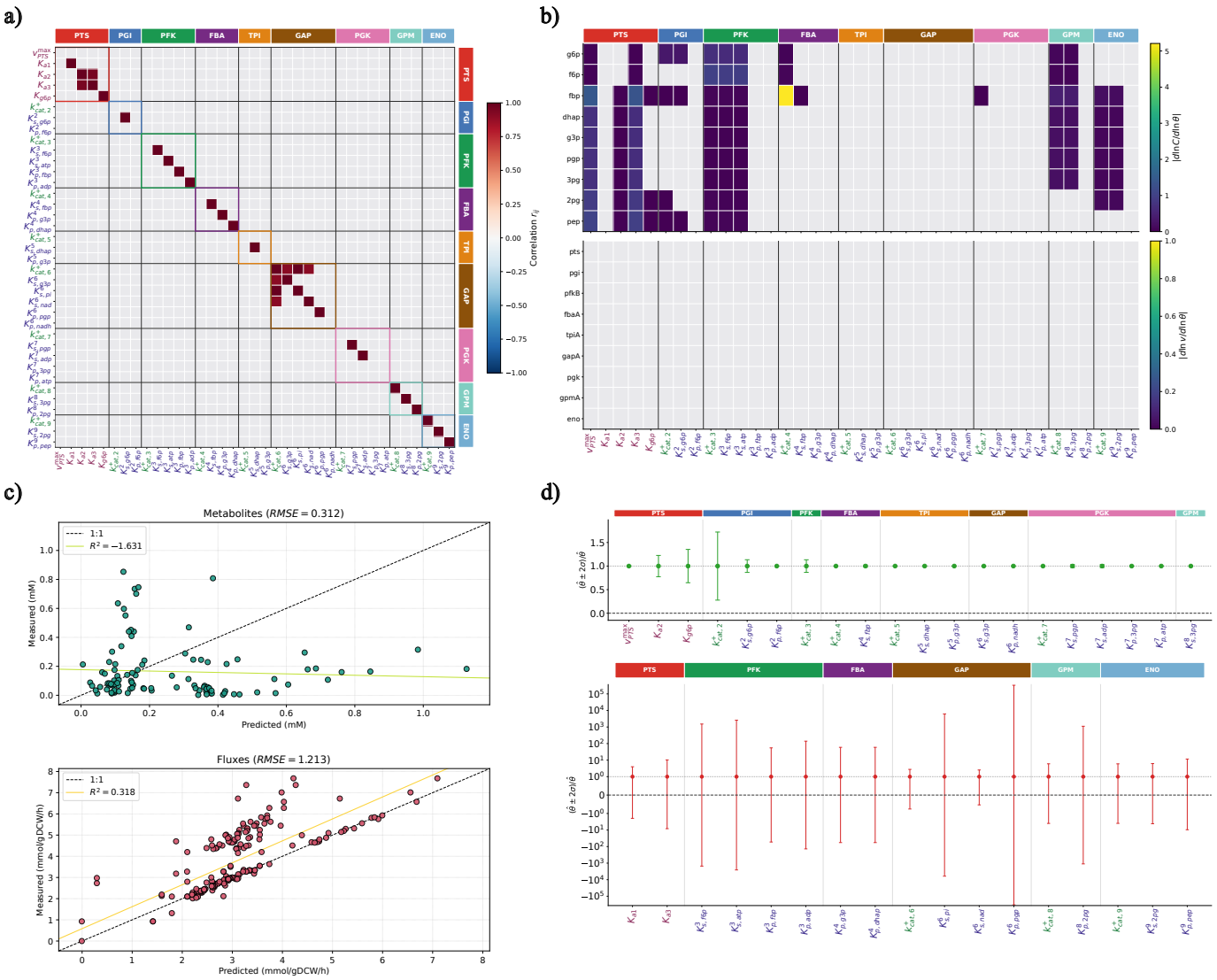


Figure S3: All-parameters (initial free) model. **a)** FIM-derived parameter correlation matrix for the 37 freely estimated parameters. **b)** Normalized sensitivity coefficients of metabolite concentrations (top, $|\partial \ln C/\partial \ln \theta|$) and fluxes (bottom, $|\partial \ln v/\partial \ln \theta|$) with respect to each parameter, grouped by reaction. **c)** Parity plots of measured vs. predicted metabolite concentrations (top, RMSE $= 0.312$) and fluxes (bottom, RMSE $= 1.213$) for the fitted 37-parameter model. **d)** FIM-derived relative confidence intervals, $(\hat{\theta} \pm 2\sigma)/\hat{\theta}$, per parameter; parameters with narrow, well-constrained intervals are shown in green (top), and those with wide or unbounded intervals in red (bottom, log scale).

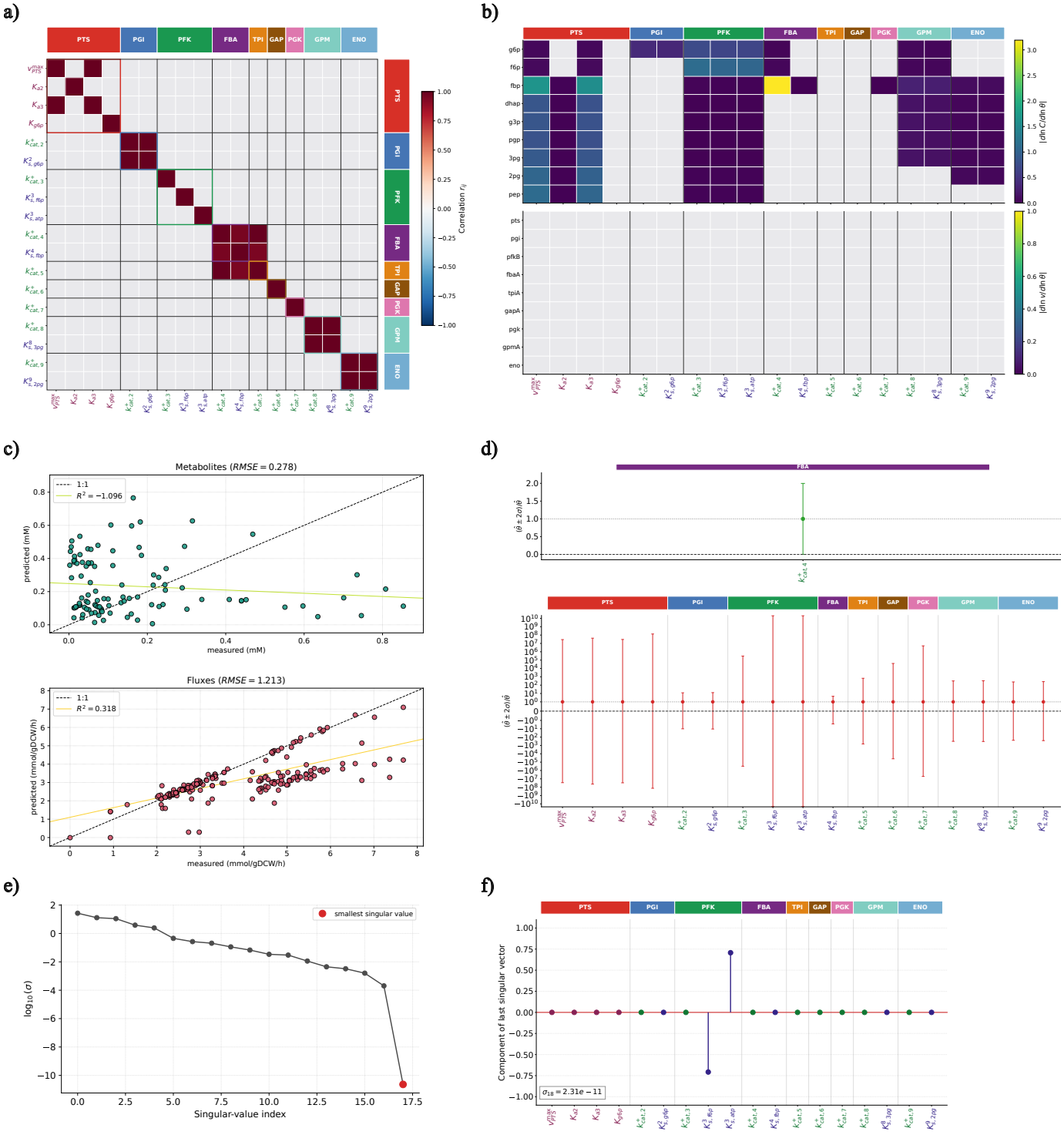


Figure S4: First reduced model (18 free parameters). **a)** FIM-derived correlation matrix after fixing 19 highly correlated, low-sensitivity parameters from the 37-parameter model (boxed blocks indicate parameters grouped by reaction). **b)** Normalized sensitivity coefficients of metabolites (top) and fluxes (bottom) for the 18 remaining free parameters. **c)** Parity plots of measured vs. predicted metabolite concentrations (top, RMSE $= 0.278$, $R^2 = -1.096$) and fluxes (bottom, RMSE $= 1.213$, $R^2 = 0.318$). **d)** FIM-derived relative confidence intervals, highlighting $k^+_{cat,4}$ (FBA) as a well-constrained parameter (top) against the full 18-parameter set (bottom, log scale). **e)** Singular value spectrum of the relative sensitivity matrix; the smallest singular value (red) reveals a near-null direction in parameter space. **f)** Contribution of each parameter to the singular vector associated with the smallest singular value, identifying $K^3_{s,f6p}$ and $K^3_{s,atp}$ (PFK) as the structurally redundant pair targeted in the second reduction step.

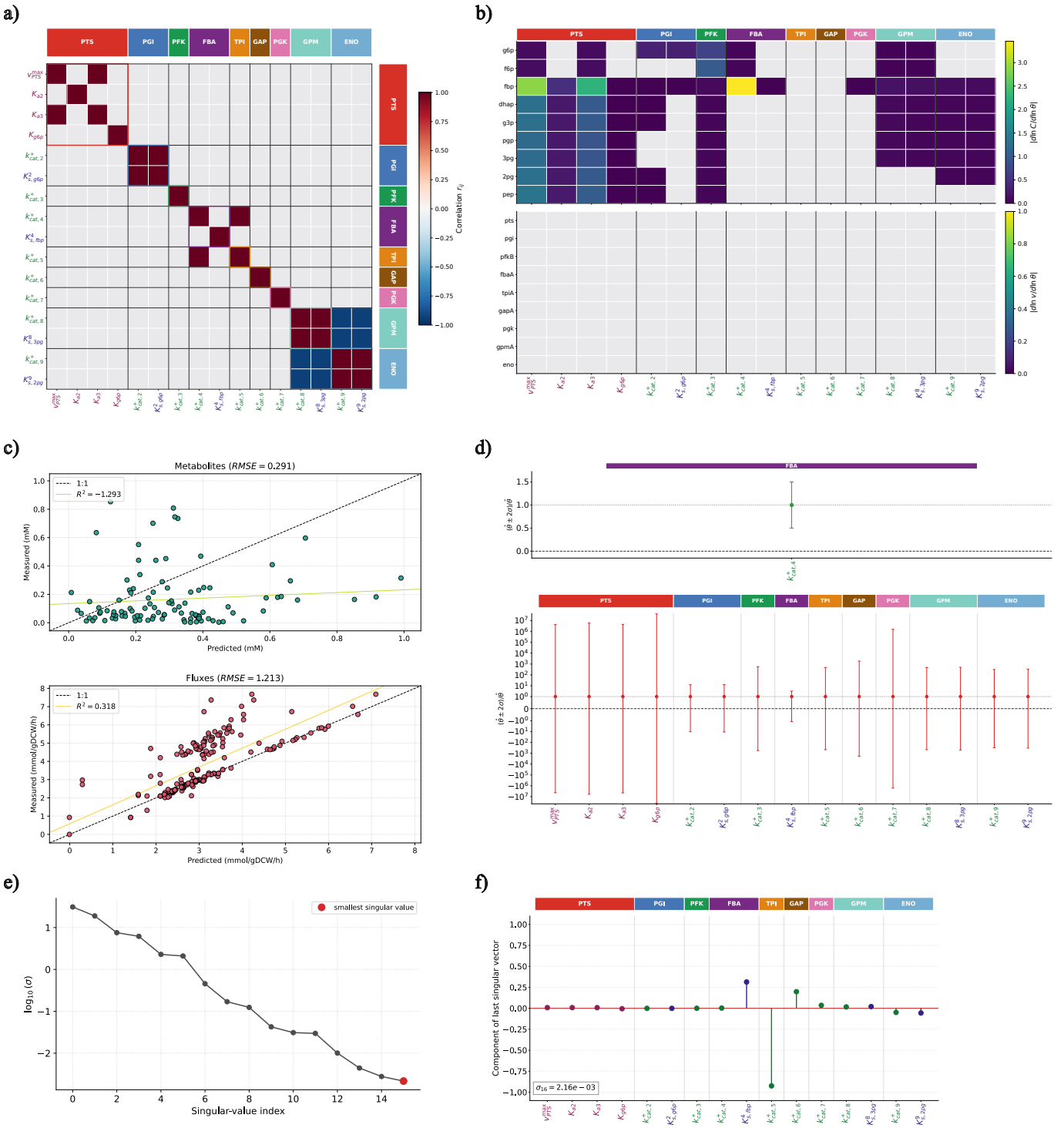


Figure S5: Final reduced model (16 free parameters). **a)** FIM-derived correlation matrix after the second reduction step. **b)** Normalized sensitivity coefficients of metabolites (top) and fluxes (bottom) for the 16 free parameters. **c)** Parity plots of measured vs. predicted metabolite concentrations (top, RMSE $= 0.291$, $R^2 = -1.293$) and fluxes (bottom, RMSE $= 1.213$, $R^2 = 0.318$). **d)** FIM-derived relative confidence intervals, highlighting $k^+_{cat,4}$ (FBA) as the most reliably constrained parameter (top) against the full parameter set (bottom, log scale). **e)** Singular value spectrum of the relative sensitivity matrix, confirming no further near-null directions beyond numerical noise ($\sigma_{16} = 2.16 \times 10^{-3}$). **f)** Contribution of each parameter to the singular vector associated with the smallest singular value, dominated by FBA and TPI parameters, consistent with the absence of additional structural redundancy.

C. Reduced model evaluation

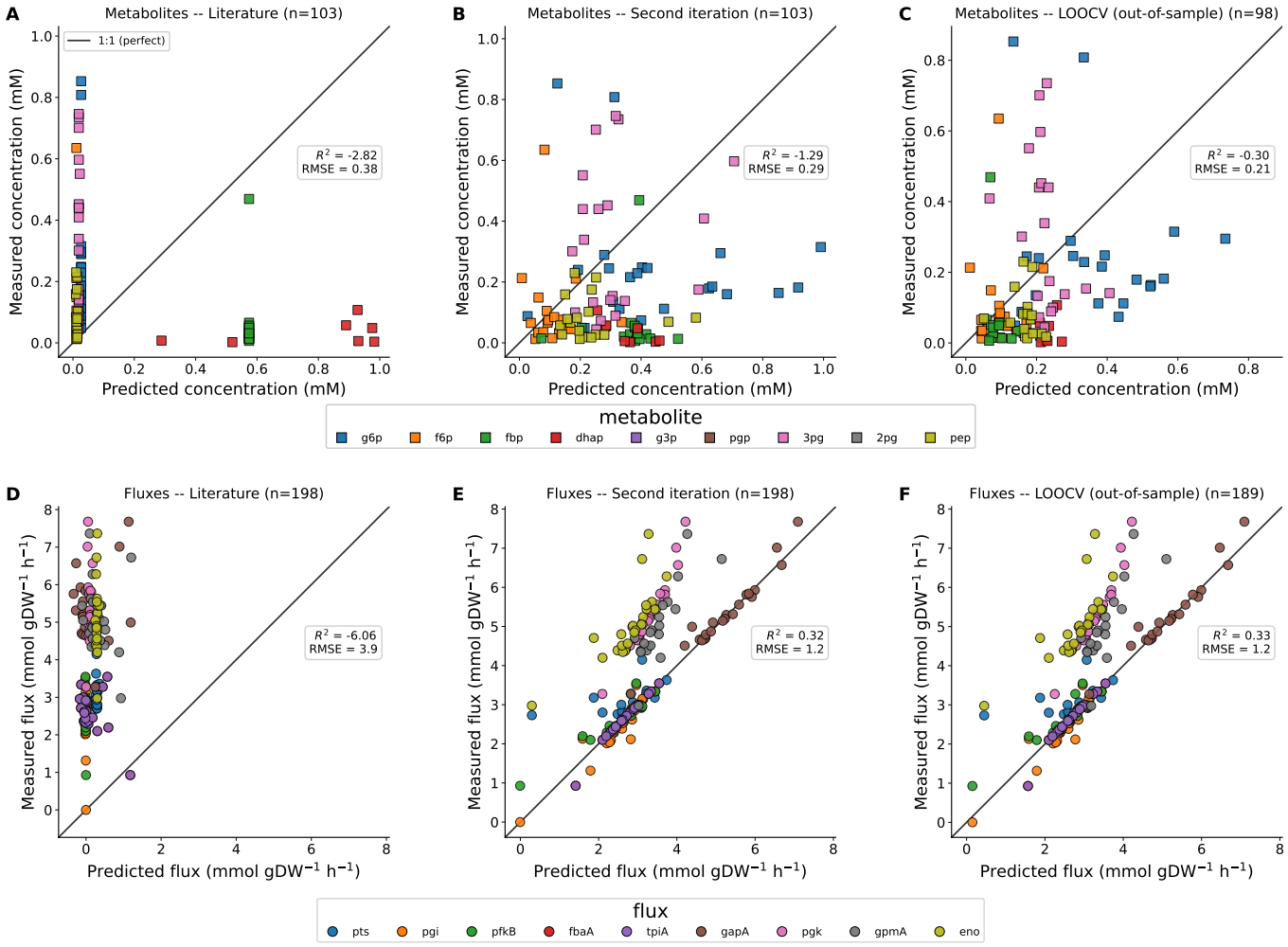


Figure S6: Comparison of model predictions across the reduction and validation pipeline. Parity plots of measured vs. predicted metabolite concentrations (**A–C**) and fluxes (**D–F**) using literature parameter values (**A, D**), the final reduced model fitted on the full training set (**B, E**), and leave-one-out cross-validation predictions on held-out conditions (**C, F**). Point colors indicate individual metabolites (**A–C**) or reactions (**D–F**).

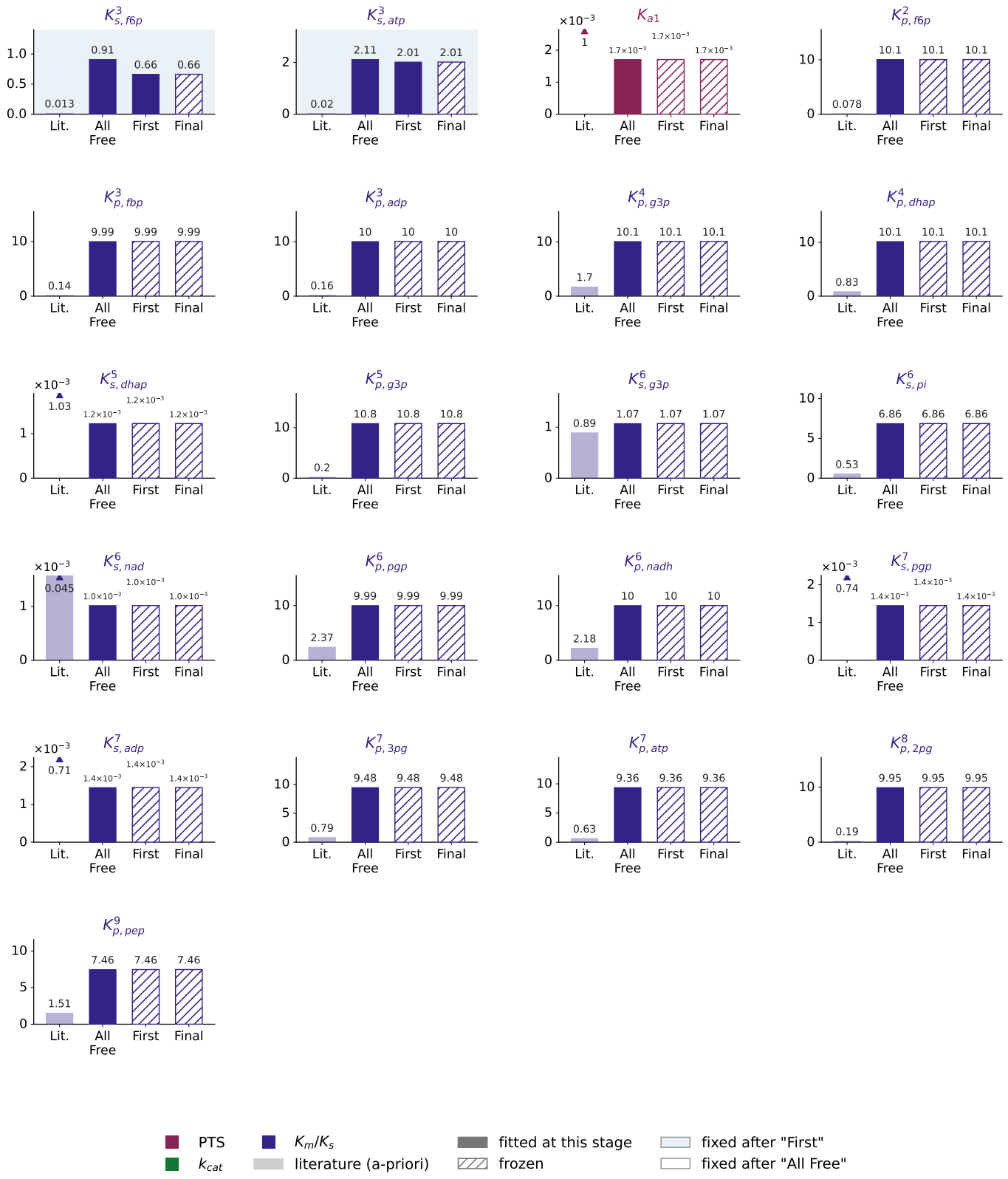


Figure S7: Evolution of parameter values across the four estimation stages: literature, initial free estimation, first reduced model, and final reduced model. Bar color indicates parameter type (PTS, k_{cat}, or K_m/K_s); fill pattern indicates the parameter's status at each stage: literature a-priori value, actively fitted, frozen at a previously fitted value, or newly fixed after the first or second reduction step.

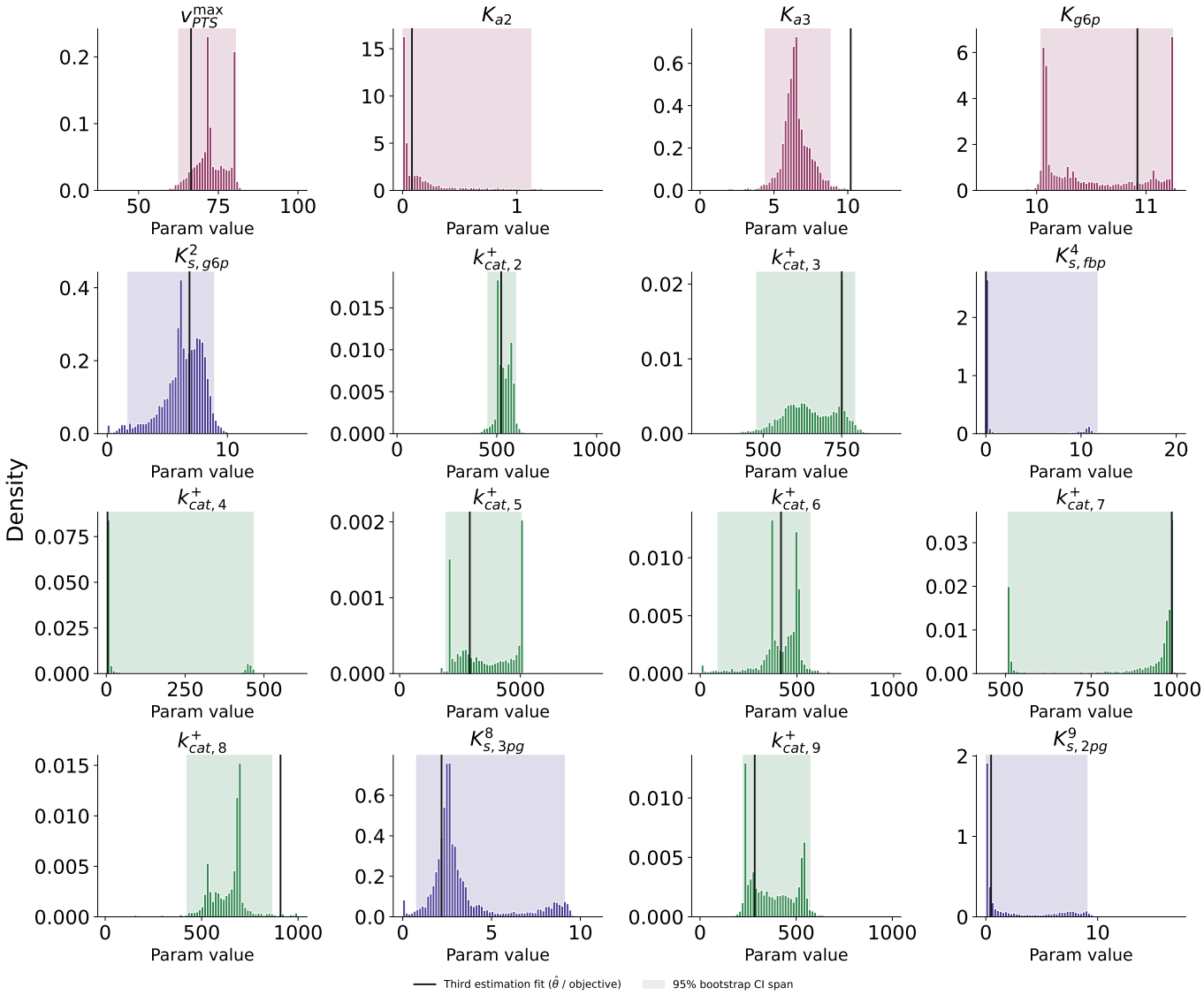


Figure S8: Bootstrap distributions ($n = 5000$) of the 16 free parameters of the final reduced model, obtained by resampling experimental conditions and re-solving the parameter estimation problem. The solid line marks the parameter estimate from the full dataset; the shaded region marks the 95% bootstrap confidence interval. Bimodal or heavy-tailed distributions indicate parameters poorly constrained by the available data.

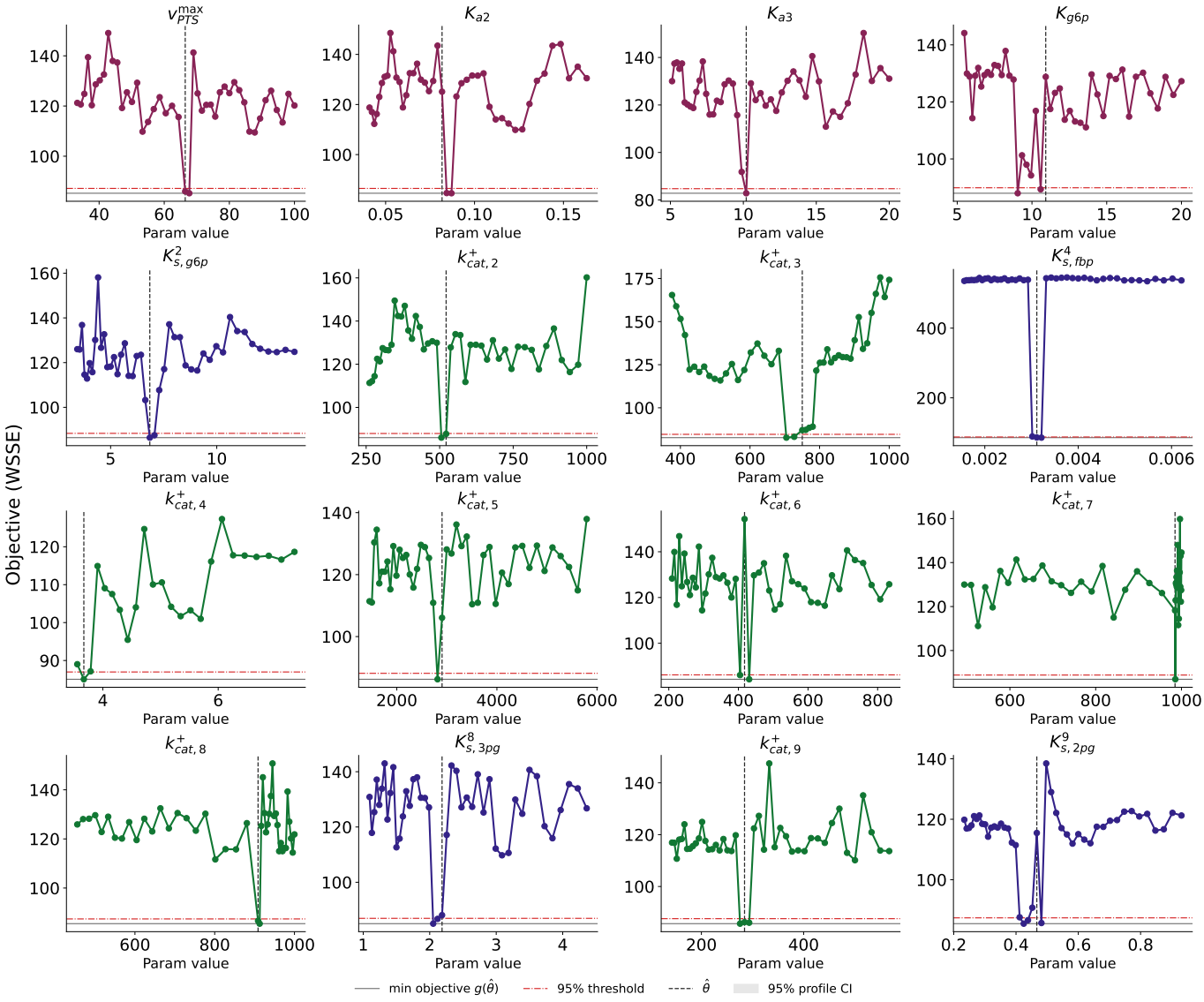


Figure S9: Profile likelihood of the 16 free parameters of the final reduced model. Each panel shows the objective function (WSSE) evaluated over a grid of fixed values for the corresponding parameter, with all remaining parameters re-optimized at each point. The dash-dotted line marks the 95% confidence threshold $\Delta(\alpha)$ above the global minimum; the dashed vertical line marks the estimated value $\hat{\theta}$. Panels with multiple points below the threshold reveal alternative local optima reached by the optimizer.

S-III. TABLES

Table I: Parameter descriptions

Symbol	Meaning
e_i	Enzyme concentration for reaction i
$k^+_{cat,i}$	Forward turnover number for reaction i
$k^r_{s,i}$	Michaelis-Menten constant for substrate i in reaction r.
$k^r_{p,i}$	Michaelis-Menten constant for product i in reaction r.
K^r_{eq}	Equilibrium constant for reaction r.
$[s]$	Substrate concentration
$[p]$	Product concentration.

Table II: Metabolites involved, abbreviation and their mass-balance classification

Abbrev.	Metabolite	Classification
glc	Glucose	Imbalanced
$g6p$	Glucose 6-phosphate	Balanced
$f6p$	Fructose 6-phosphate	Balanced
fbp	Fructose 1,6-bisphosphate	Balanced
$dhap$	Dihydroxyacetone phosphate	Balanced
$g3p$	Glyceraldehyde 3-phosphate	Balanced
pgp	1,3-Bisphosphoglycerate	Balanced
$3pg$	3-Phosphoglycerate	Balanced
$2pg$	2-Phosphoglycerate	Balanced
pep	Phosphoenolpyruvate	Balanced
pyr	Pyruvate	Imbalanced
atp	ATP	Imbalanced
adp	ADP	Imbalanced
nad	NAD^+	Imbalanced
$nadh$	NADH	Imbalanced
pi	Inorganic phosphate	Imbalanced

Table III: Glycolysis kinetic parameters in *E. coli* obtained from EcoCyc[10], Kadir et al. [12] and Equilibrator [11].

#	Enzime (gen)	Reaction	Parameters	Source
1	Phosphotransferase system	$glc + pep \rightarrow g6p + pyr$	v^{PTS}_{max}: 25.739 mmol/(gDCW·h) K_{a1}: 1.0 mM K_{a2}: 0.01 mM K_{a3}: 1.0 mM n_{g6p}: 4 K_{g6p}: 0.5 mM	[12]
2	Phosphoglucose isomerase (*Pgi*)	$g6p \rightleftharpoons f6p$	K_{eq} : 0.381 K^{g6p}_m: 1.018 mM K^{f6p}_m: 0.078 mM k_{cat} : $--$	EcoCyc: EG10702
3	Phosphofructokinase (*PfkB*)	$f6p + atp \rightarrow fbp + adp$	K_{eq} : 341.298 K^{f6p}_m: 0.013 mM K^{atp}_m: 0.02 mM K^{fbp}_m : 0.14 mM K^{adp}_m : $--$ k_{cat} : $--$	EcoCyc: EG10700

Continues on next page

Table III – continuation

#	Enzime (gen)	Reaction	Parameters	Source
4	Aldolase (*FbaA*)	$fbp \rightleftharpoons g3p + dhap$	K_{eq} : 0.1616 K_m^{fbp}: 0.24 mM K_m^{g3p} : $--$ K_m^{dhap} : $--$ k_{cat} : $10.33s^{-1} \rightarrow 0.95\frac{mmol}{mg \cdot h}$	EcoCyc: EG10282
5	TPI (*TpiA*)	$dhap \rightleftharpoons g3p$	K_{eq} : 0.113 K_m^{dhap}: 1.03 mM K_m^{g3p} : $--$ k_{cat}: 9000 $s^{-1} \rightarrow 1201.2\frac{mmol}{mg \cdot h}$	EcoCyc: EG11015
6	GAPDH (*GapA*)	$g3p + nad + pi \rightleftharpoons pgp + nadh$	K_{eq} : 0.00026 K_m^{g3p}: 0.89 mM K_m^{nad}: 0.045 mM K_m^{pi}: 0.53 mM K_m^{pgp} : $--$ K_m^{nadh} : $--$ k_{cat} : $--$	EcoCyc: EG10367
7	Phosphoglycerate kinase (*Pgk*)	$pgp + adp \rightleftharpoons 3pg + atp$	K_{eq} : 1849.572 K_m^{pgp} : $--$ K_m^{adp} : $--$ K_m^{3pg} : $--$ K_m^{atp} : $--$ k_{cat} : $--$	Equilibrator [11]
8	Phosphoglycerate mutase (*GpmA*)	$3pg \rightleftharpoons 2pg$	K_{eq} : 0.1874 K_m^{3pg}: 0.2 mM K_m^{2pg}: 0.19 mM k_{cat}: 330 $s^{-1} \rightarrow$ 41.602 $\frac{mmol}{mg \cdot h}$	EcoCyc: EG11699
9	Enolase (*Eno*)	$2pg \rightleftharpoons pep$	K_{eq} : 4.5273 K_m^{2PG}: 0.1 mM K_m^{pep} : $--$ k_{cat} : $--$	EcoCyc: EG10258

Table IV: Parameter values across estimation stages for the 37-parameter E. coli glycolysis model.

Parameter	Literature parameters	All Free	First reduced model	Final reduced model	Fixed/Free
$v_{PTS}^{\max}$	25.7	81	80.3	66.4	Free
K_{a1}	1	1.71×10^{-3}	1.71×10^{-3}	1.71×10^{-3}	Fixed
K_{a2}	0.01	0.041	0.0297	0.0817	Free
K_{a3}	1	5.3	6.47	10.2	Free
K_{g6p}	0.5	11.3	11.2	10.9	Free
$K_{s,g6p}^{2}$	1.02	6.97	6.18	6.85	Free
$K_{p,f6p}^{2}$	0.078	10.1	10.1	10.1	Fixed
$k_{cat,2}^{+}$	21.8	501	505	522	Free
$K_{s,f6p}^{3}$	0.013	0.908	0.662	0.662	Fixed
$K_{s,atp}^{3}$	0.02	2.11	2.01	2.01	Fixed
$K_{p,fbp}^{3}$	0.14	9.99	9.99	9.99	Fixed
$K_{p,adp}^{3}$	0.163	10	10	10	Fixed
$k_{cat,3}^{+}$	7.37	751	753	750	Free
$K_{s,fbp}^{4}$	0.24	1.03×10^{-3}	3.14×10^{-3}	3.11×10^{-3}	Free
$K_{p,g3p}^{4}$	1.7	10.1	10.1	10.1	Fixed
$K_{p,dhap}^{4}$	0.83	10.1	10.1	10.1	Fixed
$k_{cat,4}^{+}$	0.95	3.56	3.67	3.67	Free
$k_{cat,5}^{+}$	1.20×10^{3}	916	2.05×10^{3}	2.90×10^{3}	Free
$K_{s,dhap}^{5}$	1.03	1.23×10^{-3}	1.23×10^{-3}	1.23×10^{-3}	Fixed
$K_{p,g3p}^{5}$	0.197	10.8	10.8	10.8	Fixed
$k_{cat,6}^{+}$	77.6	348	368	418	Free
$K_{s,g3p}^{6}$	0.89	1.07	1.07	1.07	Fixed
$K_{s,pi}^{6}$	0.53	6.86	6.86	6.86	Fixed
$K_{s,nad}^{6}$	0.045	1.02×10^{-3}	1.02×10^{-3}	1.02×10^{-3}	Fixed
$K_{p,pgp}^{6}$	2.37	9.99	9.99	9.99	Fixed
$K_{p,nadh}^{6}$	2.18	10	10	10	Fixed
$k_{cat,7}^{+}$	23.9	983	984	985	Free
$K_{s,pgp}^{7}$	0.738	1.45×10^{-3}	1.45×10^{-3}	1.45×10^{-3}	Fixed
$K_{s,adp}^{7}$	0.71	1.45×10^{-3}	1.45×10^{-3}	1.45×10^{-3}	Fixed
$K_{p,3pg}^{7}$	0.792	9.48	9.48	9.48	Fixed
$K_{p,atp}^{7}$	0.632	9.36	9.36	9.36	Fixed
$k_{cat,8}^{+}$	41.6	562	694	909	Free
$K_{s,3pg}^{8}$	0.2	3.6	2.62	2.19	Free
$K_{p,2pg}^{8}$	0.19	9.95	9.95	9.95	Fixed
$k_{cat,9}^{+}$	48.8	248	237	284	Free
$K_{s,2pg}^{9}$	0.1	0.453	0.233	0.467	Free
$K_{p,pep}^{9}$	1.51	7.46	7.46	7.46	Fixed